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# LAB 09 - K-Nearest Neighbors (KNN)
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# ==============================================================

import numpy as np
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score
from scipy.stats import mode

# ==============================================================
# CUSTOM KNN CLASS (from scratch)
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class KNN_Classifier:
    def __init__(self, K=3):
        self.K = K  # Number of neighbors

    def fit(self, X_train, Y_train):
        # KNN is lazy — just store the training data
        self.X_train = X_train
        self.Y_train = Y_train
        self.m = X_train.shape[0]

    def predict(self, X_test):
        predictions = []
        for x in X_test:
            neighbors = self.get_neighbors(x)
            result = mode(neighbors, keepdims=True).mode[0]
            predictions.append(result)
        return np.array(predictions)

    def get_neighbors(self, x):
        distances = [self.euclidean(x, self.X_train[i]) for i in ran
        sorted_indices = np.argsort(distances)
        return self.Y_train[sorted_indices[:self.K]]

    def euclidean(self, x, x_train):
        # Euclidean Distance = sqrt( sum( (a - b)^2 ) )
        return np.sqrt(np.sum((x - x_train) ** 2))


# ==============================================================
# DATASET 1: STUDENT PASS/FAIL
# Features: Study Hours, Attendance %, Previous Score
# Label: Pass (1) or Fail (0)
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print("=" * 55)
print("  EXPERIMENT 1: Student Pass/Fail Dataset")
print("=" * 55)

student_data = {
    'StudyHours': [1,2,2,3,3,4,4,5,5,6,6,7,7,8,8,1,2,3,5,6],
    'Attendance': [40,50,45,55,60,65,70,75,80,85,90,88,92,95,98,35,4
    'PrevScore':  [30,40,35,50,55,60,62,70,72,78,80,85,88,90,95,25,3
    'Result':     [0, 0, 0, 0, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 0, 0
}

df_student = pd.DataFrame(student_data)
print("\nStudent Dataset (first 5 rows):")
print(df_student.head())
print(f"\nTotal students: {len(df_student)}")
print(f"Pass: {df_student['Result'].sum()} | Fail: {(df_student['Res

X  = df_student[['StudyHours','Attendance','PrevScore']].values
Y  = df_student['Result'].values
X_train, X_test, Y_train, Y_test = train_test_split(X, Y, test_size=

print(f"\nTraining samples: {len(X_train)} | Test samples: {len(X_te
print(f"\n{'K Value':<10} {'Custom KNN':<15} {'Sklearn KNN':<15}")
print("-" * 40)

for k in [1, 3, 5, 7]:
    custom = KNN_Classifier(K=k)
    custom.fit(X_train, Y_train)
    acc_custom = accuracy_score(Y_test, custom.predict(X_test)) * 10

    sk = KNeighborsClassifier(n_neighbors=k)
    sk.fit(X_train, Y_train)
    acc_sk = accuracy_score(Y_test, sk.predict(X_test)) * 100

    print(f"K = {k:<7} {acc_custom:<15.2f} {acc_sk:<15.2f}")

# Predict a new student
print("\n--- Predict a New Student ---")
new_student = np.array([[5, 80, 70]])
sk3 = KNeighborsClassifier(n_neighbors=3)
sk3.fit(X_train, Y_train)
pred = sk3.predict(new_student)
print(f"Study Hours=5, Attendance=80%, Prev Score=70")
print(f"Predicted Result: {'PASS' if pred[0]==1 else 'FAIL'}")


# ==============================================================
# DATASET 2: EMPLOYEE PERFORMANCE
# Features: Experience (years), Training Hours, Projects Done
# Label: Low(0), Medium(1), High(2)
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print("\n" + "=" * 55)
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print("  EXPERIMENT 2: Employee Performance Dataset")
print("=" * 55)

employee_data = {
    'Experience':    [1,1,2,2,3,3,4,4,5,5,6,6,7,7,8,8,9,9,10,10],
    'TrainingHours': [5,10,8,15,12,20,18,25,22,30,28,35,32,40,38,45,
    'Projects':      [1,2,2,3,3,4,4,5,5,6,6,7,7,8,8,9,9,10,10,12],
    'Performance':   [0,0,0,0,1,1,1,1,1,2,2,2,2,2,2,2,2,2,2,2]
}

df_emp = pd.DataFrame(employee_data)
print("\nEmployee Dataset (first 5 rows):")
print(df_emp.head())

X2 = df_emp[['Experience','TrainingHours','Projects']].values
Y2 = df_emp['Performance'].values
X2_train, X2_test, Y2_train, Y2_test = train_test_split(X2, Y2, test_

print(f"\nTraining samples: {len(X2_train)} | Test samples: {len(X2_
print(f"\n{'K Value':<10} {'Accuracy':<15}")
print("-" * 25)

for k in [1, 3, 5]:
    sk = KNeighborsClassifier(n_neighbors=k)
    sk.fit(X2_train, Y2_train)
    acc = accuracy_score(Y2_test, sk.predict(X2_test)) * 100
    print(f"K = {k:<7} {acc:.2f}%")

print("\n--- Predict a New Employee ---")
new_emp = np.array([[5, 30, 6]])
sk_emp = KNeighborsClassifier(n_neighbors=3)
sk_emp.fit(X2_train, Y2_train)
pred_emp = sk_emp.predict(new_emp)
labels = {0:"LOW", 1:"MEDIUM", 2:"HIGH"}
print(f"Experience=5yrs, Training=30hrs, Projects=6")
print(f"Predicted Performance: {labels[pred_emp[0]]}")


# ============================================================
# DATASET 3: FRUIT CLASSIFIER
# Features: Weight (g), Sweetness (1-10), Color Code
# Label: Apple(0), Banana(1), Orange(2)
# ============================================================

print("\n" + "=" * 55)
print("  EXPERIMENT 3: Fruit Classification Dataset")
print("=" * 55)

fruit_data = {
    'Weight':    [150,155,160,145,140,120,125,118,130,122,180,175,17
    'Sweetness': [7,6,7,8,6,9,8,9,7,8,6,5,6,7,5,7,8,6,8,6],
    'ColorCode': [1,1,1,1,1,2,2,2,2,2,3,3,3,3,3,1,1,1,2,3],
    'Fruit':     [0,0,0,0,0,1,1,1,1,1,2,2,2,2,2,0,0,0,1,2]
}
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)

df_fruit = pd.DataFrame(fruit_data)
print("\nFruit Dataset (first 5 rows):")
print(df_fruit.head())

X3 = df_fruit[['Weight','Sweetness','ColorCode']].values
Y3 = df_fruit['Fruit'].values
X3_train, X3_test, Y3_train, Y3_test = train_test_split(X3, Y3, test_

print(f"\nTraining samples: {len(X3_train)} | Test samples: {len(X3_
print(f"\n{'K Value':<10} {'Accuracy':<15}")
print("-" * 25)

for k in [1, 3, 5]:
    sk = KNeighborsClassifier(n_neighbors=k)
    sk.fit(X3_train, Y3_train)
    acc = accuracy_score(Y3_test, sk.predict(X3_test)) * 100
    print(f"K = {k:<7} {acc:.2f}%")

print("\n--- Predict a New Fruit ---")
new_fruit = np.array([[155, 7, 1]])
sk_fruit = KNeighborsClassifier(n_neighbors=3)
sk_fruit.fit(X3_train, Y3_train)
pred_fruit = sk_fruit.predict(new_fruit)
fruit_names = {0:"Apple", 1:"Banana", 2:"Orange"}
print(f"Weight=155g, Sweetness=7, Color=Red(1)")
print(f"Predicted Fruit: {fruit_names[pred_fruit[0]]}")


# ============================================================
# FINAL SUMMARY
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print("\n" + "=" * 55)
print("  FINAL COMPARISON SUMMARY (K=3)")
print("=" * 55)
print(f"{'Dataset':<30} {'Accuracy'}")
print("-" * 45)

for name, Xd, Yd in [("Student Pass/Fail", X, Y), ("Employee Perform
    Xtr, Xte, Ytr, Yte = train_test_split(Xd, Yd, test_size=0.3, ran
    sk = KNeighborsClassifier(n_neighbors=3)
    sk.fit(Xtr, Ytr)
    acc = accuracy_score(Yte, sk.predict(Xte)) * 100
    print(f"{name:<30} {acc:.2f}%")

print("\nAll experiments completed!")
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  EXPERIMENT 1: Student Pass/Fail Dataset
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Student Dataset (first 5 rows):
   StudyHours  Attendance  PrevScore  Result
0           1          40         30       0
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0          1          40          30       0
1          2          50          40       0
2          2          45          35       0
3          3          55          50       0
4          3          60          55       1

Total students: 20
Pass: 13 | Fail: 7

Training samples: 14 | Test samples: 6

K Value    Custom KNN      Sklearn KNN
----------------------------------------
K = 1      100.00          100.00
K = 3      100.00          100.00
K = 5      100.00          100.00
K = 7      33.33           33.33

--- Predict a New Student ---
Study Hours=5, Attendance=80%, Prev Score=70
Predicted Result: PASS

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  EXPERIMENT 2: Employee Performance Dataset
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Employee Dataset (first 5 rows):
   Experience  TrainingHours  Projects  Performance
0           1              5         1            0
1           1             10         2            0
2           2              8         2            0
3           2             15         3            0
4           3             12         3            1

Training samples: 14 | Test samples: 6

K Value    Accuracy
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K = 1      100.00%
K = 3      100.00%
K = 5      100.00%

--- Predict a New Employee ---
Experience=5yrs, Training=30hrs, Projects=6
Predicted Performance: HIGH

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  EXPERIMENT 3: Fruit Classification Dataset
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Fruit Dataset (first 5 rows):
   Weight  Sweetness  ColorCode  Fruit
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